



Clinical Report

SPIN4-related X-linked overgrowth in a family

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ABSTRACT

Spindlin Family Member 4 (*SPIN4*) is an epigenetic reader gene on the X chromosome. Its loss-of-function variant altering the WNT/ β -catenin pathway was recently reported to cause a *SPIN4*-associated overgrowth syndrome in an extended family.

The index case is a 14-year-old male with tall stature (+2.0 SD) as the only growth-related finding, accompanied by protruding joints, splenomegaly, low bone mineral density, and normal intelligence. Exome sequencing identified the same loss-of-function variant of *SPIN4* (NM_001012968.3:c.312_313del:p.(Arg104-Serfs*24)), which was identified in the original family. The variant was present in the proband's mother and maternal grandmother. They both had skewed X-deviation (80% and 20%) and no height gain to their mid-parental heights. The first patient with the same *SPIN4* variant described by Lui et al. had more pronounced birth weight and height compared to our patient, and an advanced bone age by one year. Both patients exhibited tall stature, normal pubertal timing, psychomotor development, and intellect, as well as similar facial features and organomegaly (Lui et al., 2023).

1. Introduction

Growth in stature is influenced by a complex interplay of genetics and/or epigenetics, hormones, nutrition, and socioeconomic conditions (Reiter and Rosenfeld, 1998). Overgrowth syndromes are a heterogeneous group of disorders primarily characterized by significant increases in height, weight, and/or head circumference, often exceeding two standard deviations from the mean on growth charts when compared to age- and sex-matched controls (Brioude et al., 2019; Tatton-Brown et al., 2017).

When overgrowth is caused by constitutional tall stature, no additional signs are usually present. In cases where other clinical problems are also observed, epigenetic and/or genetic factors might be responsible for the overgrowth (Brioude et al., 2019). Some examples are dysmorphic facial features, macrocephaly, developmental delays, intellectual disabilities (e.g., in Sotos and Weaver syndromes), and increased susceptibility to malignancies (e.g., Beckwith-Wiedemann syndrome) (Manor and Lalani, 2020). Therefore, accurate diagnosis is

crucial for effective management of comorbidities of overgrowth-associated conditions.

Numerous genetic alterations can promote linear bone growth, including pathogenic variants in genes involved in epigenetic regulation. The most well-known of these are Weaver syndrome and Sotos syndrome, caused by pathogenic variants in the histone methyltransferases *EZH2* and *NSD1*, respectively (Ocansey and Tatton-Brown, 1993; Tatton-Brown et al., 1993), and Tatton-Brown-Rahman syndrome, caused by pathogenic variants in the DNA methyltransferase *DNMT3A* (Ostrowski and Tatton-Brown, 1993). These three syndromes are mainly characterized by tall stature, macrocephaly, and mild to severe intellectual disability. In 2023, a new overgrowth syndrome caused by a defect in the epigenetic regulator gene *SPIN4* (NM_001012968.3, c.312_313AGdel p. (Arg104Serfs*24)) was identified (Lui et al., 2023).

SPIN4-associated overgrowth syndrome is a generalized overgrowth disorder of prenatal onset characterized by extremely tall stature, enlarged liver and spleen, and macrocephaly, but with normal intelligence in the affected male proband. The first male patient with the

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hemizygous variant showed more profound enhanced growth than female family members carrying the heterozygous variant. Compared to other known and previously mentioned overgrowth syndromes, the first patient with *SPIN4*-associated overgrowth syndrome exhibited normal psychomotor development and intellect during childhood (Lui et al., 2023).

This article describes the second male patient with *SPIN4*-associated overgrowth syndrome caused by the same genetic defect of *SPIN4* gene identified in Estonia to expand the genotype-phenotype spectrum.

2. Clinical report

The proband (Fig. 2, IV-1), a 14-year-old boy, was referred to the Competence Centre for Rare Diseases to evaluate the clinical significance of a pathogenic frameshift variant (NM_001012968.3, c.312_313del:p. (Arg104Serfs*24)) in the *SPIN4* gene. Since tall stature is relatively common in the local population, tall stature alone was not the reason for genetic testing. The patient's primary complaints were exercise intolerance and distal lower extremity fatigue. Due to a family history of McArdle's disease, genetic testing was performed, which incidentally identified a pathogenic loss-of-function variant in the *SPIN4* gene.

The proband was born large for gestational age, with a birth weight of 4980 g (+3.0 SD) and length of 55 cm (+1.5 SD). Standard deviation (SD) scores are applied to height measurements using growth charts based on local population data (Grünberg et al., 1998). The mother had no history of gestational diabetes. He exhibited normal psychomotor development and intelligence and has had no issues with attending the regular school program.

The patient has been growing tall, with a height of 179 cm (+2.0 SD), a weight of 58 kg (+0.75 SD), and a head circumference of 58 cm (+1.5 SD) at the age of 14 (Grünberg et al., 1998). On physical exam, he exhibited protruding joints, mild genu varum, and inward-turned ankles and feet (Fig. 1A). Neurological examination was unremarkable, showing no signs of muscular weakness or hypotonia. He had characteristic facial features previously described, including down-slanting, deep-set eyes, and a prominent philtrum.

Assessment of puberty revealed Tanner stage 5 (Genitalia 5; Pubic hair stage 5) (Marshall and Tanner, 1970), with testicular volume >25ml at the age of 14. Bone age, estimated at 13.99 years by automated

analysis (BoneXpert 3.0.3) (Thodberg et al., 2009), corresponded to chronological age. However, manual analysis was 13 years 6 months (Z-score -0.58), giving 90.2% of height completion based on the Bayley-Pinneau method, therefore predicted adult height of 198 cm (Table 1; Fig. 1B). The initial study showed WNT/ β -catenin suppression by *SPIN4* knockout (*Spin4* KO) in mice as the underlying mechanism of bone growth. The impact of the WNT/ β -catenin state was studied in bone cells (Lui et al., 2023). Therefore, we measured dual-energy X-ray absorptiometry and found overall low bone mineral density (BMD) in the Z-score of whole body less the head (Table 1).

Clinical biochemical evaluations, including iCa (1.27 mmol/L), P (1.62 mmol/L), Mg, and creatinine, were in the normal range. TSH was 2.24 (0.51-4.30 mU/L), free thyroxine (T4) 12.2 (12.6-21.0 pmol/L), PTH 2.97 (1.6-7.2 pmol/L), and Vitamin D (25-OH) 63.0 nmol/L. His IGF-1 was 363.0 μ g/L (115-489 μ g/L) and IGFBP-3 6.8 mg/L (3.0-10.0 mg/L), both within the normal range. Abdominal ultrasound found splenomegaly, which was asymptomatic at the time of evaluation and showed no signs of anaemia. Brain magnetic resonance imaging was normal.

The family history on the mother's side raised suspicion of inheriting the *SPIN4* truncating variant from her side of the family (Fig. 2). The proband's mother had a height of 172 cm with her mid-parental height of 168 cm (Fig. 2, III-3), and the maternal grandmother's height was 166.8 cm with her mid-parental height of 176 cm (Fig. 2, II-2). The maternal grandmother's father was a tall man with a height of 200 cm (Fig. 2, I-2), as was the son of her father's sister (Fig. 2, II-3).

According to the proband's mid-parental height calculations (father's height: 175 cm), the target height of the proband is 180 cm. Based on his bone age of 13 years 6 months, his predicted adult height is approximately 198 cm (+2.75 SDS). Therefore, carrying *SPIN4* KO added +2.75 height SDS to his genetic potential. Based on his bone age of 14 years, the predicted adult height is approximately 195.8 cm (+2.64 SDS).

3. Methods and results

Whole exome sequencing (WES) was performed as part of the clinical evaluation. Genomic DNA was extracted from whole blood. Libraries were prepared using the DNA Prep with Exome 2.5 Enrichment Kit (coding region 37.5 Mb) and sequenced on an Illumina NovaSeq X Plus platform (2x151bp paired-end reads). Secondary analysis of the raw

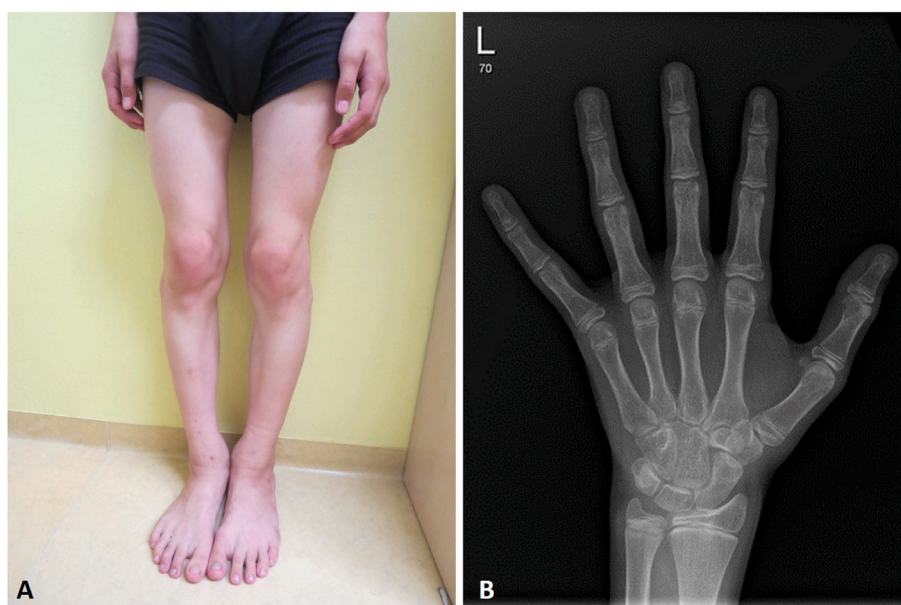


Fig. 1. (A) Lower extremities at age 14, showing prominent knees and ankles; (B) Proband's bone age X-ray (chronological age 14.1 years; the bone age read as 13 years, 6 months).

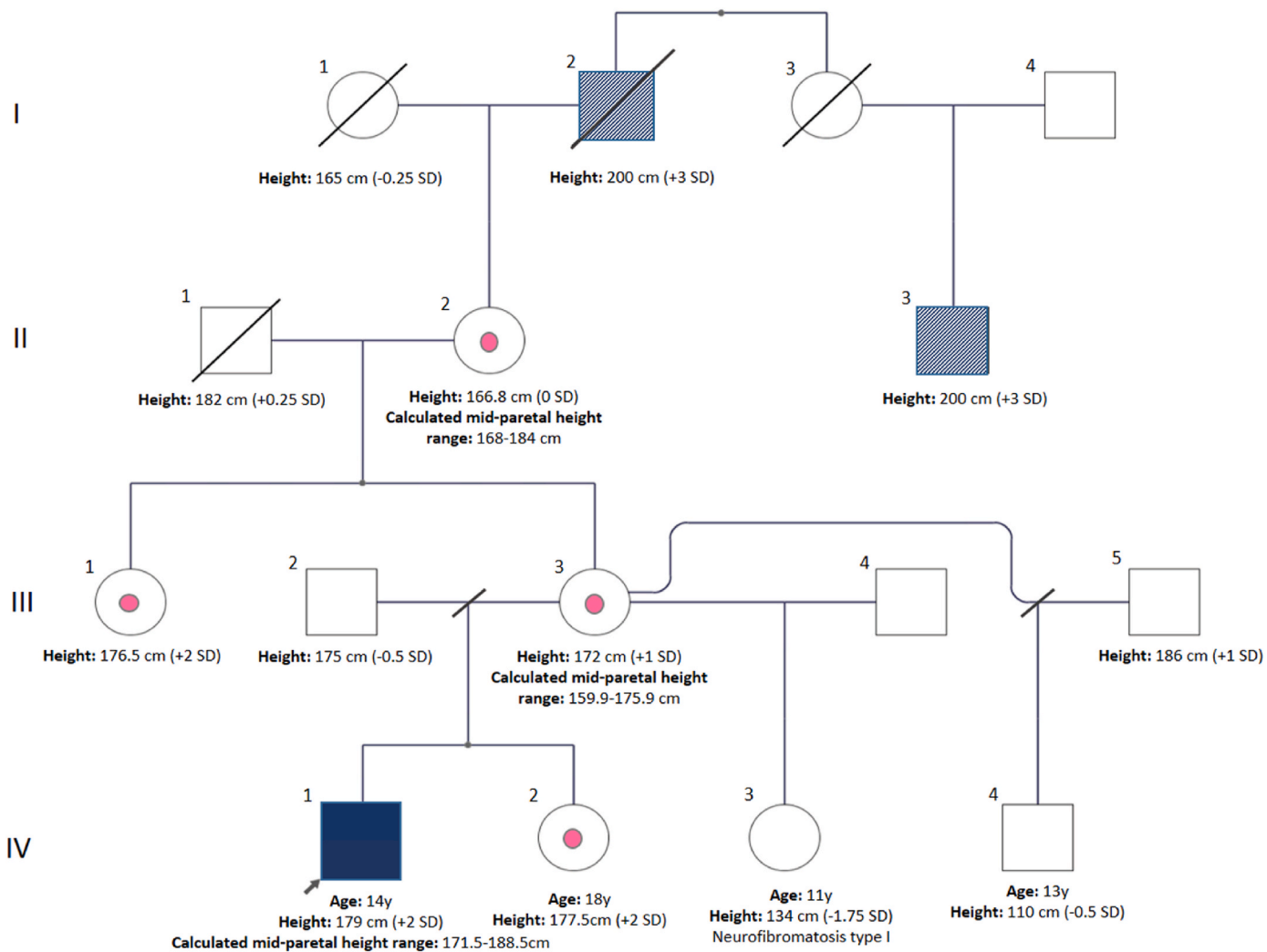


Fig. 2. Pedigree of the family. The proband is indicated by an arrow and a blue closed rectangle. The proband's sister, mother, mother's sister, and maternal grandmother are *SPIN4* carriers. The proband's half-sister (IV-3) and half-brother (IV-4) are tested; they do not carry the pathogenic *SPIN4* variant and have no tall stature. The proband's maternal grandmother's father (I-2) and the son of her father's sister (II-3) could be *SPIN4* carriers based on their heights, but they are unavailable for testing. Standard deviation (SD) scores are applied to height measurements using growth charts based on our population data (Grünberg et al., 1998).

Table 1
Bone age and bone mineral density measurements.

Bone age measurements		
Chronological age	14.09 years	
Greulich-Pyle	13.99 years, Z-score 0.16	
Tanner-Whitehouse-3 (TW3)	14.03 years	
Bone Health Index (BHI) ^a	4.22 (−2.34 SD)	
Bone mineral density measurements		
	Unadjusted Z-Score	Height-Adjusted Z-Score (CHOP BMD Pediatric Calculator) ^b
L1-L4 BMD	+0.19 SD (Zemel et al., 2011)	−1.08 SD
TBLH BMD ^c	−2.03 SD	−3.29 SD

^a The Bone Health Index (BHI), calculated by the BoneXpert software, is based on measurements of the cortical thickness of metacarpals II–IV.

^b CHOP BMD Pediatric Calculator is an online tool developed by the Children's Hospital of Philadelphia (CHOP, <https://zscore.research.chop.edu/calcpedbonedens.php>).

^c TBLH – total body less head; BMD – bone mineral density.

sequencing data was performed with DRAGEN software v4.1.23 using

Illumina DRAGEN Enrichment Application (v1.2.1). Sequence reads were mapped and aligned to the hg19 reference genome. The resulting genomic variants were annotated using VEP (v113). WES identified a previously described pathogenic hemizygous frameshift variant in the *SPIN4* gene (NM_001012968.3, c.312_313del:p.(Arg104Serfs*24)). The variant was confirmed by Sanger sequencing, and familial segregation analysis identified the variant in the proband's mother and maternal grandmother. The variant was not found in the general population data (gnomAD v4.1).

To exclude skewed X chromosome inactivation, we analysed the androgen receptor (AR) gene CAG repeat region of the patient's mother and maternal grandmother using fluorescent-labelled PCR amplification, digested with *RsaI* and *HpaII* restriction enzymes. PCR fragments were detected using Applied Biosystems GeneMapper 6 Software on an ABI 3500XL capillary sequencer. We found unequal X-inactivation in the mother (20/80%) and maternal grandmother (22/78%). However, on which X chromosome the variant is located was not studied.

4. Discussion

Our report presents the second case of *SPIN4*-associated overgrowth syndrome. Our findings provide further evidence of the clinical phenotype in male carriers of the pathogenic *SPIN4* variant causing tall

stature.

SPIN4 is an epigenetic reader gene which function was predicted to bind to modified histone peptides based on the three Tudor domains of *SPIN4*. It is assumed that *SPIN4* primarily acts as an epigenetic reader, but each domain's precise role has not yet been established. The initial study demonstrated that *SPIN4* normally binds specific modified histone peptides, promotes canonical WNT signalling, and inhibits cell proliferation *in vitro*, and the identified frameshift variant lost all these functions in primary chondrocytes. The *SPIN4* mouse models with truncating variants demonstrated the impact of hemizygous male and heterozygous and homozygous female mice on large body size and enhanced bone growth compared to the wild type, respectively. The examination of mouse model growth plates revealed increased cell proliferation in the proliferative zone and an increased number of progenitor chondrocytes in the resting zone, likely induced by suppressed WNT/ β -catenin activation, which is likely the underlying mechanism of enhanced bone growth (Lui et al., 2023).

Our proband exhibited similar symptoms to those observed in the original patient with the *SPIN4* variant. The first patient with the same *SPIN4* variant described by Lui et al. had more pronounced birth weight and height (+4.3 SDS and +4.8 SDS, respectively) compared to our patient, and an advanced bone age by one year. Both patients exhibited tall stature, normal pubertal timing, psychomotor development, and intellect, as well as similar facial features and organomegaly (Lui et al., 2023).

Although *SPIN4*-associated overgrowth syndrome shared some phenotypes (such as tall stature) with other overgrowth syndromes, some differences, including normal psychomotor development and normal intellect, were also noted. Future epigenetic investigations of the *SPIN4* gene could potentially inform novel therapeutic strategies for individuals presenting with short stature and preserved cognitive function.

Sotos, Weaver, and Tatton-Brown syndromes typically present with mild to severe developmental delays and learning disabilities (Ocansey and Tatton-Brown, 1993; Ostrowski and Tatton-Brown, 1993; Tatton-Brown et al., 1993). The underlying cause of intellectual disability in these syndromes is not well known, but causative genes might significantly affect neuronal function. So far, the original patient and our patient have not shown cognitive defects or learning difficulties. However, this requires confirmation from a larger cohort of patients with *SPIN4*-associated overgrowth syndrome. No clinical manifestations of adult male patients are currently available. Therefore, a future nature history study for the complete phenotype across age would be required.

Epigenetic overgrowth syndromes can present with advanced bone age, especially in Sotos syndrome, which reduces the total period for linear growth during childhood, resulting in a normal height in adulthood (Lui and Baron, 2024). In our case, bone age was slightly delayed or close to his chronological age. Because the dynamics of bone age advancement and the exact timing of epiphyseal fusion remain unknown in *SPIN4* loss-of-function, monitoring his growth until final adult height is reached will provide valuable longitudinal insight into skeletal maturation. Conventional height prediction methods, such as Bayley-Pinneau, should be interpreted cautiously in this context because their accuracy may be reduced in cases with atypical pubertal progression or delayed epiphyseal closure.

The previous mouse and human study investigated the modulation of WNT/ β -catenin by *Spin4* KO (Lui et al., 2023). Low WNT/ β -catenin activation by genetic defects is a well-established cause of low bone mineral density or osteoporosis (Hu et al., 2024). Because no data is available to show that *Spin4* KO also induces WNT/ β -catenin suppression in bone marrow mesenchymal stem cells or osteoblasts, we measured the patient's bone mineral density. Patients' whole body less head bone mineral density was significantly low. However, our patient had no history of fractures. Given his normal physical activity and normal PTH/vitamin D levels, we cautiously speculate that a cause of low bone mineral density might be due to a pathogenic variant in the

SPIN4 gene. The decreased free T4 level may have influenced bone metabolism and skeletal maturation, potentially contributing to the lack of advanced bone age observed in this patient.

Hemizygous males are expected to have a more severe phenotype than heterozygous females, a pattern typical for X-linked conditions. Our proband's mother and maternal grandmother were *SPIN4* truncating variant carriers but had normal height and did not exceed their expected mid-parental height range. Due to this difference in previously described female carriers, we hypothesized that our proband's mother and maternal grandmother might have skewed X-inactivation, with the normal *SPIN4* allele being more pronounced. We analysed androgen receptor (AR)-CAG trinucleotide repeat length in the proband, mother, and maternal grandmother to test this hypothesis. We found unequal X-inactivation in the mother (20/80%) and maternal grandmother (22/78%). Although we cannot determine which allele contains the loss-of-function *SPIN4* variant, we can speculate that both individuals might have a more pronounced normal *SPIN4* expression, and this might explain why female carriers did not gain height benefit (additional adult height to their genetic potential) from the frameshift *SPIN4* variant. However, since height is a complex trait highly susceptible to environmental and other genetic interactions, skewed X-chromosome inactivation alone is insufficient to explain the phenotype. Potential tissue-specific variations in X-chromosome inactivation must also be considered, for example, between peripheral blood and growth plates. In addition, *SPIN4* may play a dose-independent role in regulating normal growth processes.

Genetic anomalies and pathways involved in overgrowth syndromes, such as abnormalities in DNA methylation and histone tail modifications, are often observed in tumours. This overlap may increase the risk of developing tumours in individuals with these syndromes (Brioude et al., 2019). However, research has shown that the types of variants, particularly in genes related to epigenetic regulation, differ between overgrowth syndromes and cancer. The reasons for these differences are complex and may include the embryonic lethality of certain oncogenic mutations when they occur in the germline (Tatton-Brown et al., 2017). In the case of Sotos and Weaver syndromes, there have been occasional reports of tumours, including neuroblastoma, teratomas, and acute leukemia (Ocansey and Tatton-Brown, 1993; Tatton-Brown et al., 1993). In a population-based study of seven million children (Connolly et al., 2024), children with overgrowth syndromes and overgrowth features showed a significantly increased incidence of cancers when presenting with hepatosplenomegaly. However, which specific genetic defects would be associated with which type of cancer requires further investigation. Therefore, although we understand the risk, specific tumour surveillance is not established. In our case, there was no family history of cancer. Further studies are needed to determine whether *SPIN4*-related overgrowth syndrome is associated with increased cancer risk.

5. Summary

This report presents the second documented case of X-linked overgrowth syndrome caused by a loss-of-function variant in the *SPIN4* gene. Unlike other epigenetic overgrowth syndromes, our patient demonstrated normal intellectual development similar to the previously described *SPIN4*-associated overgrowth syndrome. Our patient showed low bone mineral density with no history of fractures. The proband's mother and maternal grandmother were heterozygous *SPIN4* truncating variant carriers, but did not exceed their expected mid-parental height. Further cases are needed to clarify the phenotypic spectrum associated with *SPIN4* variants in males and females and long-term health risks, including cancers.

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CRediT authorship contribution statement

Lisanna Põlluaas: Conceptualization, Investigation, Visualization, Writing – original draft. **Stella Lilles:** Investigation, Writing – review & editing. **Aleksandr Peet:** Investigation, Writing – review & editing. **Youn Hee Jee:** Supervision, Writing – review & editing. **Ülle Murumets:** Investigation, Writing – review & editing. **Mihkel Ilisson:** Writing – review & editing. **Mare Lintrop:** Investigation, Writing – review & editing. **Katrin Õunap:** Funding acquisition, Investigation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Katrin Õunap reports financial support was provided by Estonian Research Council grant PRG2040. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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